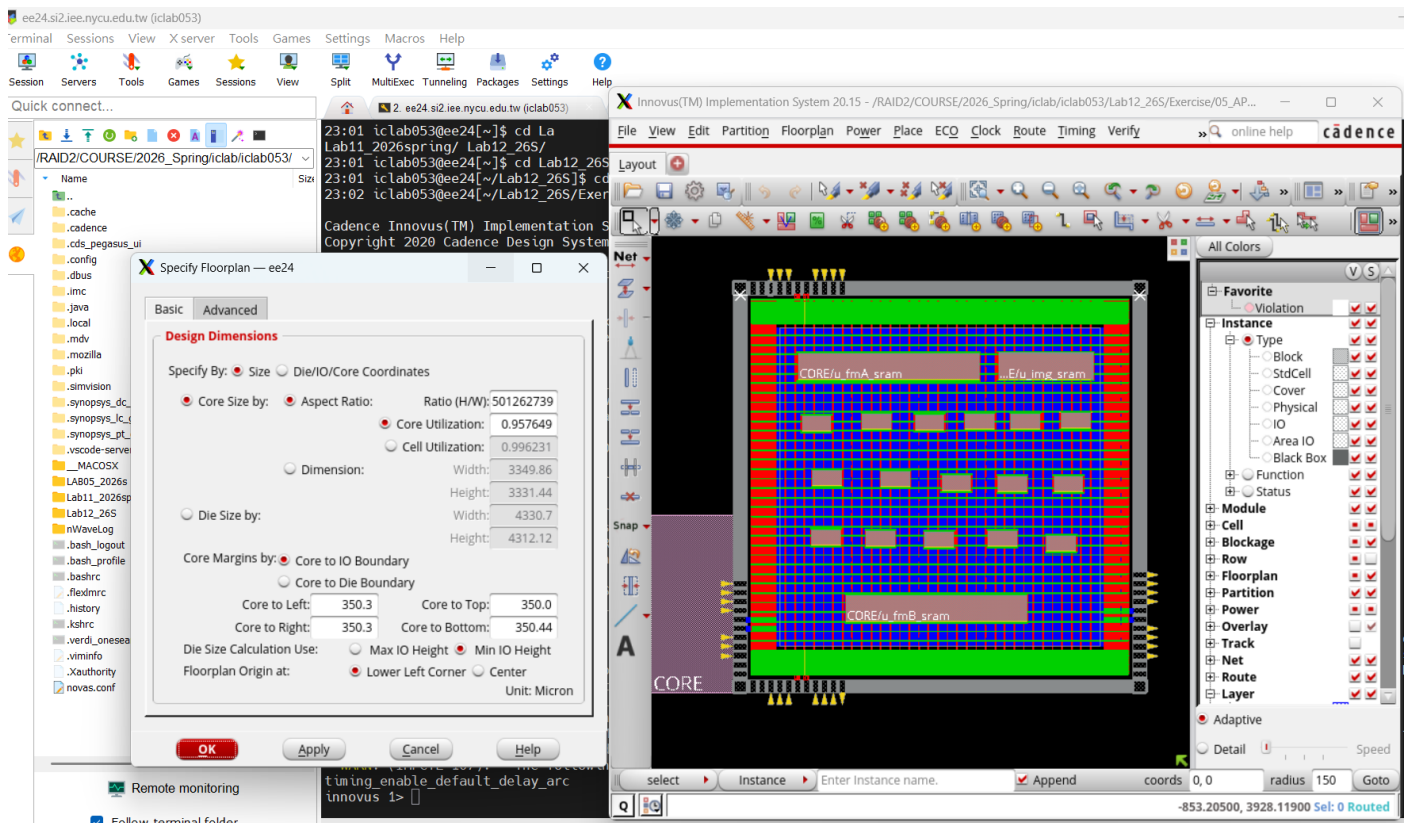
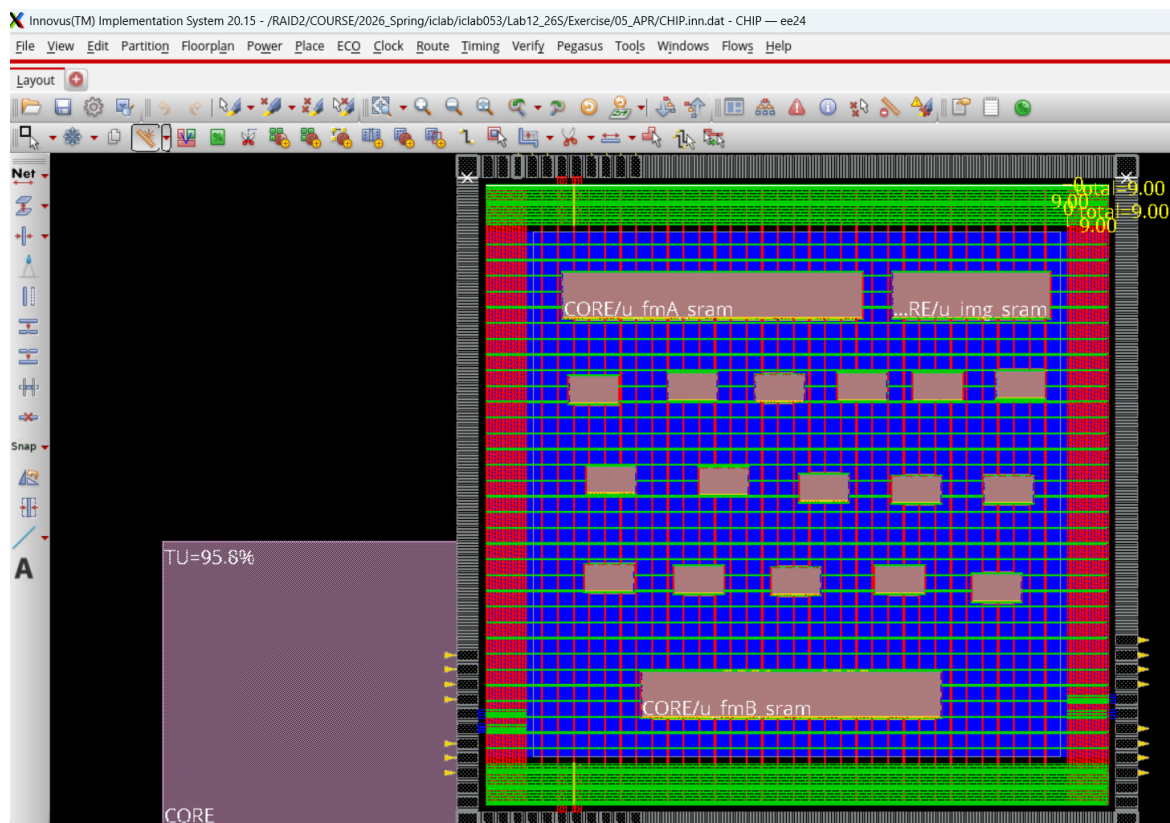
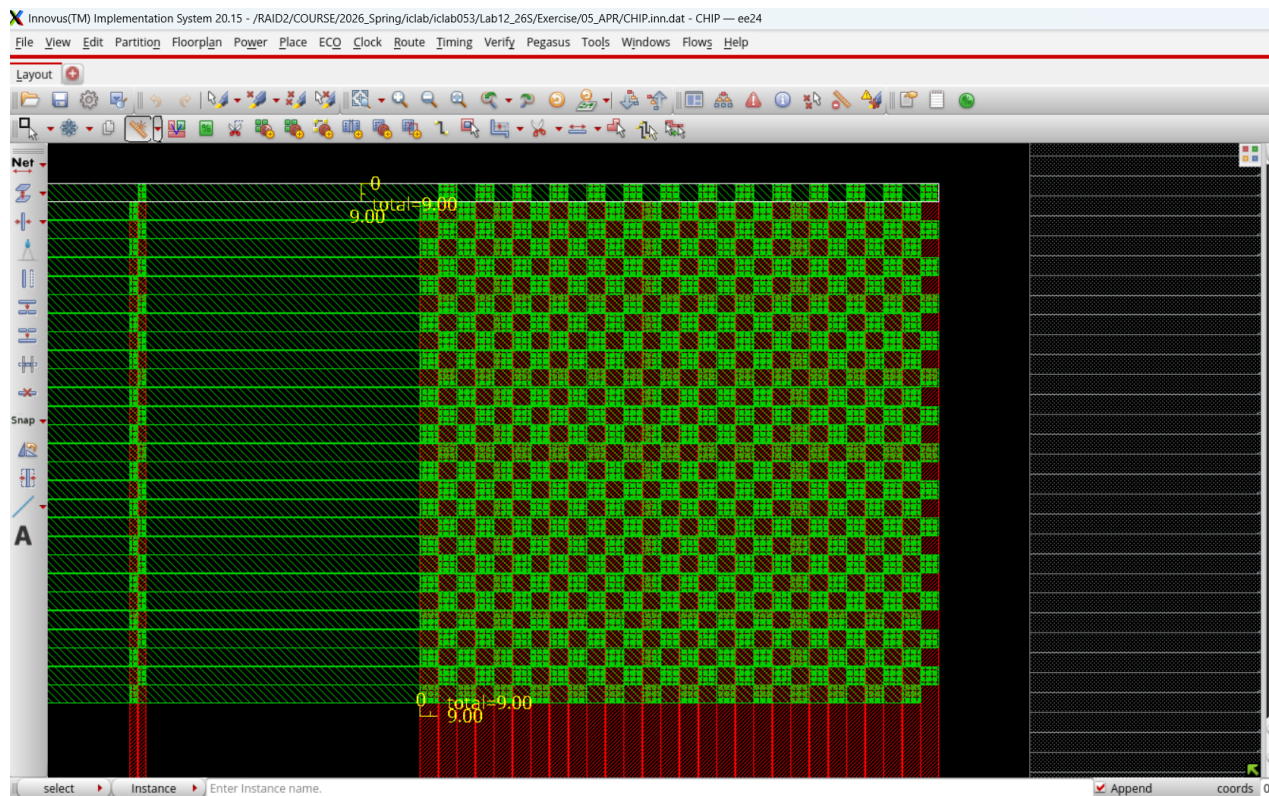


# Report

## 1. Core to IO boundary :



## 2. Core Ring :



### 3. Post-Route setup time analysis:

The screenshot shows the MobaXterm interface with a terminal window displaying the output of a 'timeDesign' command. The terminal output includes a summary of setup views, timing data for various parameters, and overall project statistics.

**timeDesign Summary**

Setup views included:  
av\_func\_mode\_max

| Setup mode       | all   | reg2reg | default |
|------------------|-------|---------|---------|
| WNS (ns):        | 0.625 | 0.625   | 2.332   |
| TNS (ns):        | 0.000 | 0.000   | 0.000   |
| Violating Paths: | 0     | 0       | 0       |
| All Paths:       | 4144  | 4013    | 599     |

| DRVs       | Real           |           | Total          |
|------------|----------------|-----------|----------------|
|            | Nr nets(terms) | Worst Vio | Nr nets(terms) |
| max_cap    | 0 (0)          | 0.000     | 0 (0)          |
| max_tran   | 0 (0)          | 0.000     | 0 (0)          |
| max_fanout | 0 (0)          | 0         | 0 (0)          |
| max_length | 0 (0)          | 0         | 0 (0)          |

Density: 15.256%  
(126.528% with Fillers)  
Total number of glitch violations: 0

Reported timing to dir timingReports  
Total CPU time: 94.39 sec  
Total Real time: 96.0 sec  
Total Memory Usage: 2391.199219 Mbytes  
Reset AAE Options  
\*\*\* timeDesign #1 [finish] : cpu/real = 0:01:34.4/0:01:36.0 (1.0), totSession cpu/real = 0:02:04.2/0:04:42.3 (0.4), mem = 2391.2M  
innovus 1>

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

## 4. Post-Route hold time analysis :

The screenshot shows a MobaXterm terminal window with a file explorer on the left and a terminal output on the right. The terminal output displays the results of a post-route hold time analysis performed using the 'fullDC' (full delay calculation) method. The analysis includes a summary of hold views, a table of hold times for various paths, and a density calculation.

Terminal Output:

```
Start delay calculation (fullDC) (1 T). (MEM=2377.2)
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Skipped = 0.
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Analyzed = 43864.
Total number of fetched objects 43864
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 43959, 0.1 percent of the nets selected for SI analysis
End delay calculation. (MEM=2419.37 CPU=0:00:00.2 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2419.37 CPU=0:00:00.3 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:13.5 real=0:00:13.0 totSessionCpu=0:02:24 mem=2419.4M)
```

timeDesign Summary

Hold views included:  
av\_func\_mode\_min

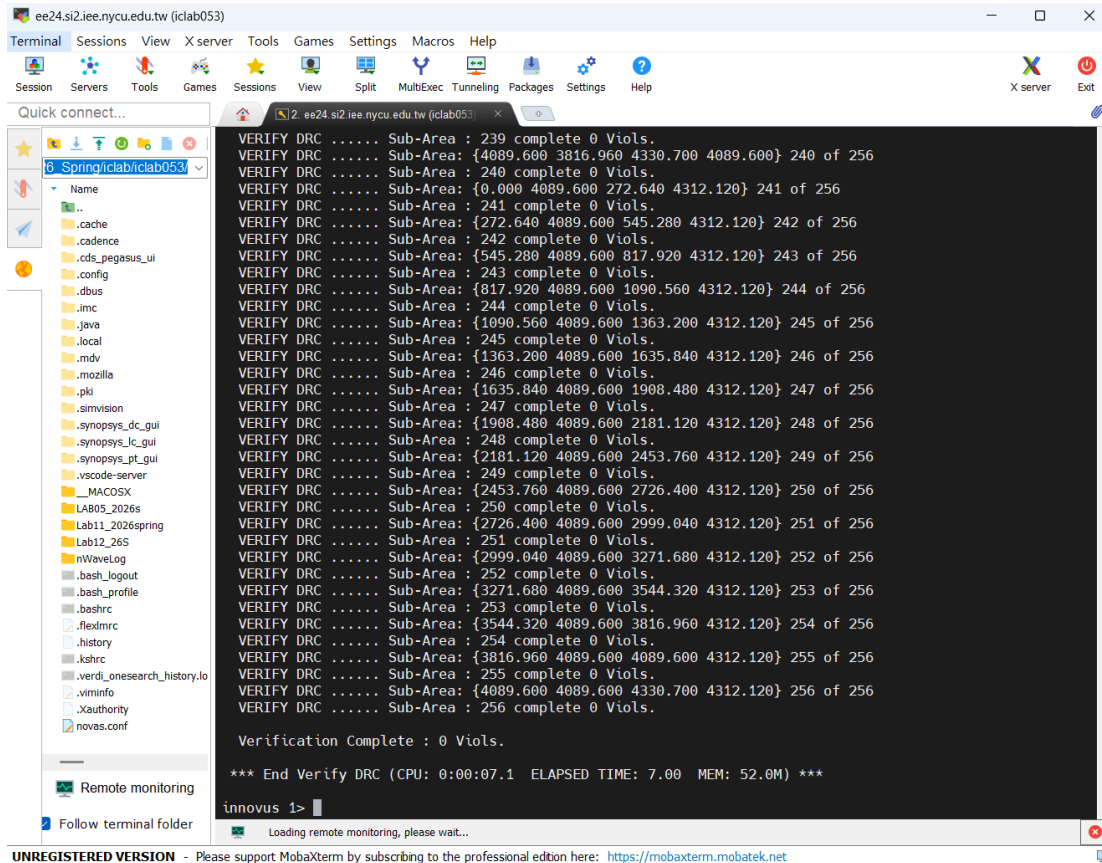
| Hold mode        | all   | reg2reg | default |
|------------------|-------|---------|---------|
| WNS (ns):        | 0.091 | 0.091   | 9.528   |
| TNS (ns):        | 0.000 | 0.000   | 0.000   |
| Violating Paths: | 0     | 0       | 0       |
| All Paths:       | 4144  | 4013    | 599     |

Density: 15.256%  
(126.528% with Fillers)

Reported timing to dir timingReports  
Total CPU time: 16.4 sec  
Total Real time: 17.0 sec  
Total Memory Usage: 2334.632812 Mbytes  
Reset AAE Options  
\*\*\* timeDesign #2 [finish] : cpu/real = 0:00:16.4/0:00:16.7 (1.0), totSession cpu/real = 0:02:24.0/0:06:13.4 (0.4), mem = 2334.6M  
innovus 1>

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

## 5. DRC result :



The screenshot shows a MobaXterm window with a terminal session. The left sidebar displays a file tree for the project 'Spring/iciab/iciab053'. The terminal output shows the results of a Design Rule Check (DRC) for the 'Spring' project. The results indicate that all sub-areas passed the DRC with 0 violations. The verification was completed at 13:49:56 on May 28, 2026, with a CPU time of 0:00:07.1, an elapsed time of 7.00, and a memory usage of 52.0M.

```
ee24.si2.lee.nycu.edu.tw (iciab053)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
6 Spring/iciab/iciab053
Name
.cache
.cadence
.cds_pegasus_ui
.config
.dbus
.imc
.java
.local
.mdv
.mozilla
.pki
.simvision
.synopsys_dc_gui
.synopsys_ic_gui
.synopsys_pt_gui
.vscodeserver
__MACOSX
LAB05_2026s
Lab11_2026spring
Lab12_26S
nWaveLog
.bash_logout
.bash_profile
.bashrc
.flexmrc
.history
.kshrc
.verdi_oneseach_history.lo
.viminfo
.xauthority
.novas.conf

Remote monitoring
Follow terminal folder

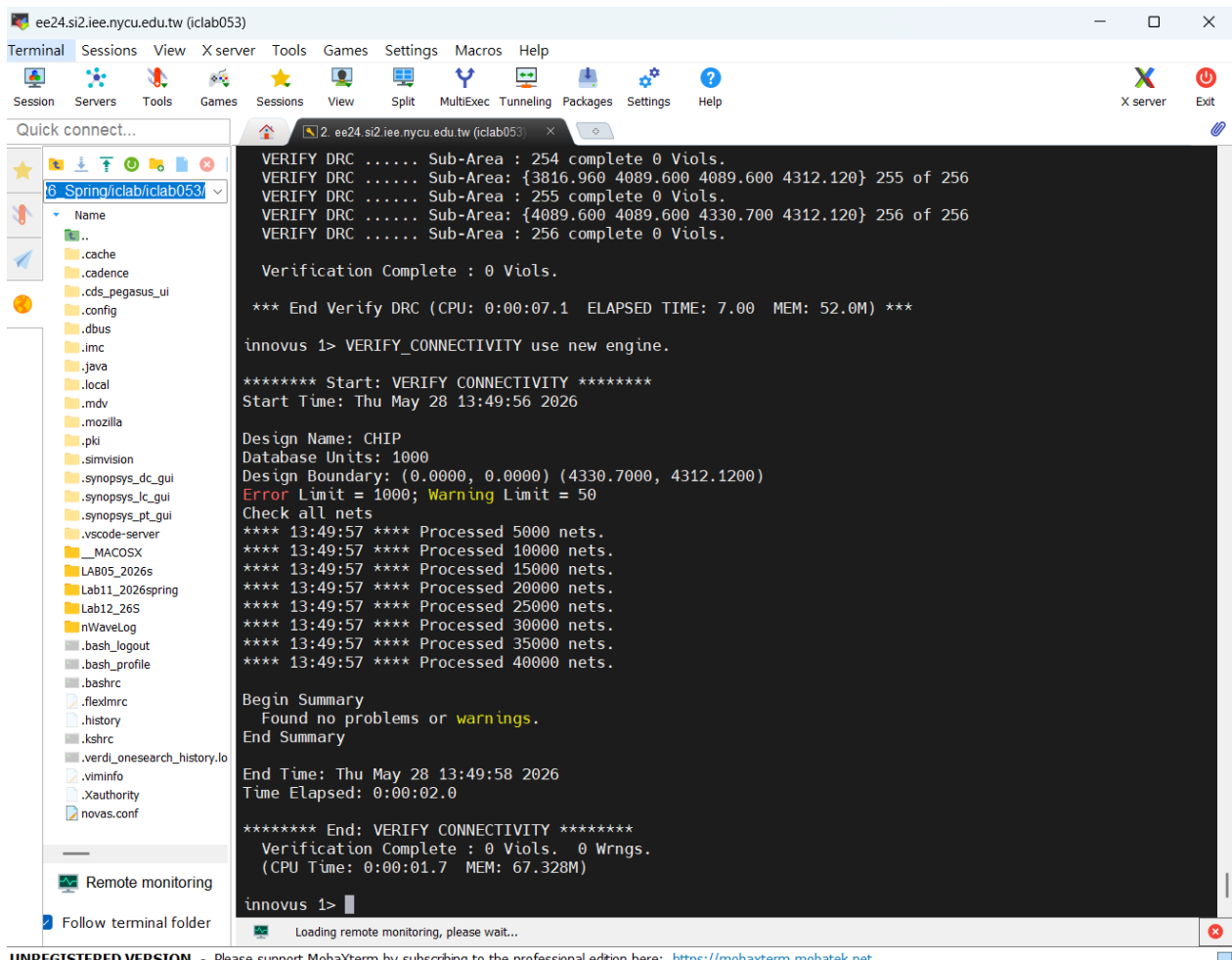
ee24.si2.lee.nycu.edu.tw (iciab053)
VERIFY DRC ..... Sub-Area : 239 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {4089.600 3816.960 4330.700 4089.600} 240 of 256
VERIFY DRC ..... Sub-Area : 240 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 4089.600 272.640 4312.120} 241 of 256
VERIFY DRC ..... Sub-Area : 241 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {272.640 4089.600 545.280 4312.120} 242 of 256
VERIFY DRC ..... Sub-Area : 242 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {545.280 4089.600 817.920 4312.120} 243 of 256
VERIFY DRC ..... Sub-Area : 243 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {817.920 4089.600 1090.560 4312.120} 244 of 256
VERIFY DRC ..... Sub-Area : 244 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1090.560 4089.600 1363.200 4312.120} 245 of 256
VERIFY DRC ..... Sub-Area : 245 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1363.200 4089.600 1635.840 4312.120} 246 of 256
VERIFY DRC ..... Sub-Area : 246 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1635.840 4089.600 1908.480 4312.120} 247 of 256
VERIFY DRC ..... Sub-Area : 247 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1908.480 4089.600 2181.120 4312.120} 248 of 256
VERIFY DRC ..... Sub-Area : 248 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2181.120 4089.600 2453.760 4312.120} 249 of 256
VERIFY DRC ..... Sub-Area : 249 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2453.760 4089.600 2726.400 4312.120} 250 of 256
VERIFY DRC ..... Sub-Area : 250 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2726.400 4089.600 2999.040 4312.120} 251 of 256
VERIFY DRC ..... Sub-Area : 251 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2999.040 4089.600 3271.680 4312.120} 252 of 256
VERIFY DRC ..... Sub-Area : 252 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3271.680 4089.600 3544.320 4312.120} 253 of 256
VERIFY DRC ..... Sub-Area : 253 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3544.320 4089.600 3816.960 4312.120} 254 of 256
VERIFY DRC ..... Sub-Area : 254 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3816.960 4089.600 4089.600 4312.120} 255 of 256
VERIFY DRC ..... Sub-Area : 255 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {4089.600 4089.600 4330.700 4312.120} 256 of 256
VERIFY DRC ..... Sub-Area : 256 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:07.1 ELAPSED TIME: 7.00 MEM: 52.0M) ***

innovus 1>
```

## 6. LVS result :



The screenshot shows a MobaXterm window with a terminal session. The left sidebar displays a file tree for the project 'Spring/iciab/iciab053'. The terminal output shows the results of a Logical Verification Step (LVS) for the 'Spring' project. The results indicate that all sub-areas passed the LVS with 0 violations. The verification was completed at 13:49:58 on May 28, 2026, with a CPU time of 0:00:01.7, an elapsed time of 0:00:02.0, and a memory usage of 67.328M.

```
ee24.si2.lee.nycu.edu.tw (iciab053)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
6 Spring/iciab/iciab053
Name
.cache
.cadence
.cds_pegasus_ui
.config
.dbus
.imc
.java
.local
.mdv
.mozilla
.pki
.simvision
.synopsys_dc_gui
.synopsys_ic_gui
.synopsys_pt_gui
.vscodeserver
__MACOSX
LAB05_2026s
Lab11_2026spring
Lab12_26S
nWaveLog
.bash_logout
.bash_profile
.bashrc
.flexmrc
.history
.kshrc
.verdi_oneseach_history.lo
.viminfo
.xauthority
.novas.conf

Remote monitoring
Follow terminal folder

ee24.si2.lee.nycu.edu.tw (iciab053)
VERIFY DRC ..... Sub-Area : 254 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {3816.960 4089.600 4089.600 4312.120} 255 of 256
VERIFY DRC ..... Sub-Area : 255 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {4089.600 4089.600 4330.700 4312.120} 256 of 256
VERIFY DRC ..... Sub-Area : 256 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:07.1 ELAPSED TIME: 7.00 MEM: 52.0M) ***

innovus 1> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Thu May 28 13:49:56 2026

Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (4330.7000, 4312.1200)
Error Limit = 1000; Warning Limit = 50
Check all nets
**** 13:49:57 **** Processed 5000 nets.
**** 13:49:57 **** Processed 10000 nets.
**** 13:49:57 **** Processed 15000 nets.
**** 13:49:57 **** Processed 20000 nets.
**** 13:49:57 **** Processed 25000 nets.
**** 13:49:57 **** Processed 30000 nets.
**** 13:49:57 **** Processed 35000 nets.
**** 13:49:57 **** Processed 40000 nets.

Begin Summary
Found no problems or warnings.
End Summary

End Time: Thu May 28 13:49:58 2026
Time Elapsed: 0:00:02.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:01.7 MEM: 67.328M)

innovus 1>
```



## 7. Post Layout simulation result :

ee24.si2.tee.nycu.edu.tw (iclab053)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/RAID2/COURSE/2026\_Spring/iclab/iclab053/

Name Size

- ..
- stdcells.cl
- technonly.cl
- .top.power.db
- .top.power\_1.db
- .top.power\_2.db
- cellIDMap
- CHIP.rpt
- CHIP\_GND.pp
- CHIP\_VCC.pp
- CHIP\_VDD.pp
- power.db
- power.db.cnstr.tcl
- power\_1.db.cnstr.tcl
- power\_2.db.cnstr.tcl
- static\_default\_ground\_rail.ptiavg
- static\_default\_power\_rail.ptiavg
- static\_GND.ptiavg
- static\_VCC.ptiavg
- stdcells.report
- stdcells.summary
- technonly.report
- technonly.summary
- voltus\_power.ptifiles

```
make[1]: Leaving directory `./RAID2/COURSE/2026_Spring/iclab/iclab053/Lab12_26S/Exercise/06_POST/csrc'
Chronologic VCS simulator copyright 1991-2022
Contains Synopsys proprietary information.
Compiler version T-2022.06_Full64; Runtime version T-2022.06_Full64; May 28 14:15 2026
Doing SDF annotation ..... Done
RESET START
*Verdi* Loading libsscore_vcs201906.so
FSDB Dumper for VCS, Release Verdi_P-2019.06, Linux x86_64/64bit, 05/26/2019
(C) 1996 - 2019 by Synopsys, Inc.
*****
* ERROR -
* The simulator version is newer than the FSDB dumper version which
* may cause abnormal behavior, please contact Synopsys support for
* assistance.
*****
*Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the
programs that are using this file.
*Verdi* : Create FSDB file 'CHIP.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : End of traversing.
RESET OVER
WEIGHT INPUT START
WEIGHT INPUT OVER
START PATTERN 0
IMG INPUT OVER
PASS PATTERN 0 WITH 31045 CYCLES
=====
ALL PATTERN PASS !!!
Average latency: 31045 cycles
Total latency: 31045 cycles
=====
$finish called from file "PATTERN.v", line 107.
$finish at simulation time 811170000 ps
VCS Simulation Report
Time: 811170000 ps
CPU Time: 319.310 seconds; Data structure size: 13.2Mb
Thu May 28 14:18:54 2026
CPU time: 5.538 seconds to compile + 1.786 seconds to elab + .917 seconds to link + 319.3
96 seconds in simulation
14:18 iclab053@ee24[~/Lab12_26S/Exercise/06_POST]$
```

Remote monitoring

☒ Follow terminal folder

Loading remote monitoring, please wait...

## 8. Power result :

ee24.si2.tee.nycu.edu.tw (iclab053)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...

/RAID2/COURSE/2026\_Spring/iclab/iclab053/

Name Size

- stdcells.cl
- techonly.cl
- .top.power.db 4
- .top.power\_1.db 4
- .top.power\_2.db 4
- cellIDMap 10
- CHIP.rpt 29
- CHIP\_GND.pp 3
- CHIP\_VCC.pp 3
- CHIP\_VDD.pp 3
- power.db 13
- power.db.cnstr.tcl 1
- power\_1.db.cnstr.tcl 1
- power\_2.db.cnstr.tcl 1
- static\_default\_ground\_rail.ptiavg 4
- static\_default\_power\_rail.ptiavg 4
- static\_GND.ptiavg 891
- static\_VCC.ptiavg 891
- stdcells.report 165
- stdcells.summary 21
- techonly.report 147
- techonly.summary 16
- voltus\_power.ptifiles 1

Remote monitoring

Follow terminal folder

```
Ended Power Analysis: (cpu=0:05:58, real=0:07:01, mem(process/total/peak)=2370.43MB/6406.80MB/7049.10MB)

Begin Boundary Leakage Calculation
Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2370.43MB/6406.80MB/7049.10MB)
Begin Static Power Report Generation
16 instances have no static power
** WARN: (VOLTUS_POWER-2152): Instance VDDP0 (VCC3IOD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance GNDP0 (GNDIOD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance VDDP1 (VCC3IOD) has no static power.
** WARN: (VOLTUS_POWER-2152): Instance GNDP1 (GNDIOD) has no static power.
** WARN: (EMS-27): Message (VOLTUS_POWER-2152) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.

*

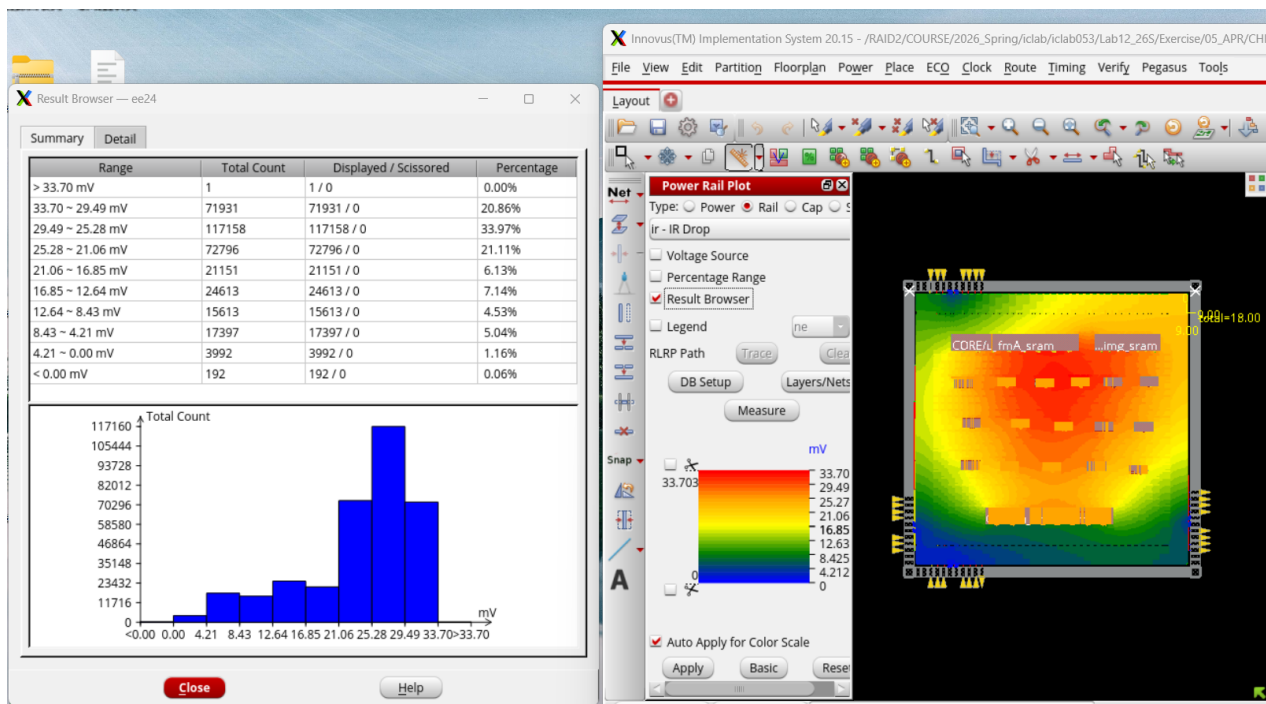
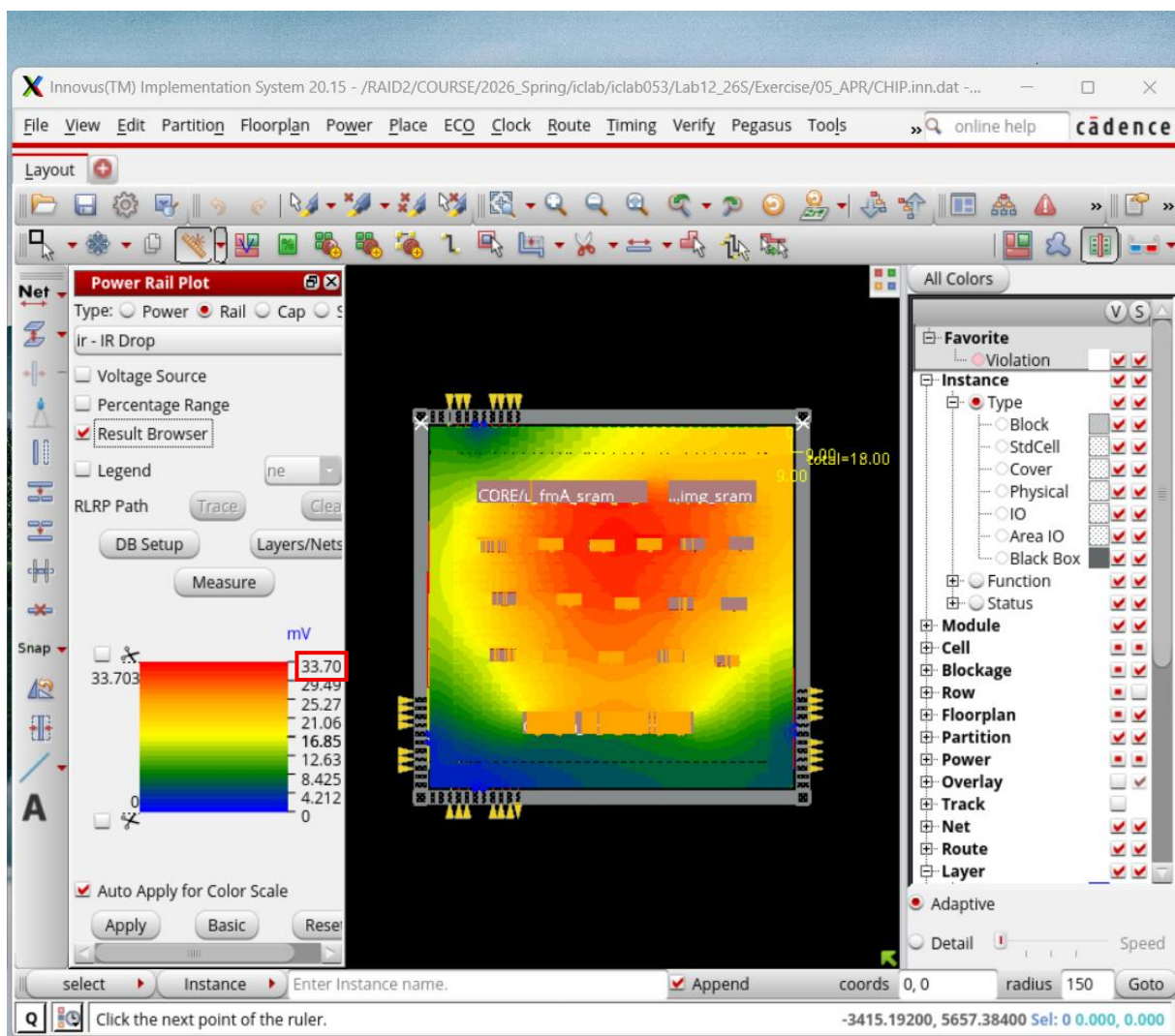
Total Power
-----
Total Internal Power:      156.74474626      87.3302%
Total Switching Power:    22.39213347      12.4758%
Total Leakage Power:       0.34829883       0.1941%
Total Power:              179.48517857

-----
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2392.95MB/6422.80MB/7049.10MB)

Begin Creating Binary Database
Ended Creating Binary Database: (cpu=0:00:01, real=0:00:01, mem(process/total/peak)=2906.60MB/7193.23MB/7049.10MB)

Output file is power_log/CHIP.rpt
```

## 9. IR Drop Results :



我的IR drop在第一次做時約為37mV，再稍微改過power pad位置以及增加stripes數量後就稍微下降到33mV了，後來翻看學長做的一些筆記發現把晶片比例調成方形，以及增加decoupling cell似乎有效，不過能達到50mV的spec就沒有多做嘗試了。